No. of Printed Pages : 4

BCS-012

BACHELOR OF COMPUTER APPLICATIONS (BCA) (Revised)

Term-End Examination

June, 2021

BCS-012 : BASIC MATHEMATICS

Time : 3 hours *Maximum Marks : 100*

Note : *Question number **1** is **compulsory**. Attempt any **three** questions from the remaining questions.*

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1. (a) If $A = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}$; $B = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$ and $(A + B)^2 = A^2 + B^2$, find a and b. *5*

(b) If the first term of an AP is 22, the common difference is – 4, and the sum to n terms is 64, find n. *5*

(c) Find the angle between the lines

$$\vec{r}_1 = 2\hat{i} + 3\hat{j} - 4\hat{k} + t(\hat{i} - 2\hat{j} + 2\hat{k})$$

$$\vec{r}_2 = 3\hat{i} - 5\hat{k} + s(3\hat{i} - 2\hat{j} + 6\hat{k}).$$ *5*

(d) If α, β are roots of $x^2 - 2kx + k^2 - 1 = 0$, and $\alpha^2 + \beta^2 = 10$, find k. *5*

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(e) If $y = 1 + ln\,(x + \sqrt{x^2 + 1})$, prove that

$$(x^2 + 1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 0.$$ 5

(f) Find the points of discontinuity of the following function : 5

$$f(x) = \begin{cases} x^2, & x > 0 \\ x + 3, & x \le 0 \end{cases}$$

(g) Solve the inequality $\frac{5}{|x - 3|} < 7.$ 5

(h) Evaluate the integral

$$I = \int \frac{x^2}{(1 + x)^3}\,dx.$$ 5

2. (a) Use the principle of mathematical induction to show that

$2 + 2^2 + \ldots + 2^n = 2^{n+1} - 2$ for each natural number n. 5

(b) Using determinant, find the area of the triangle whose vertices are (1, 2); (– 2, 3) and (– 3, – 4). 5

(c) Draw the graph of the solution set for the following inequalities :

$$2x + y \ge 8,\ x + 2y \ge 8 \text{ and } x + y \le 6$$ 5

(d) Use De Moivre's theorem to find $(i + \sqrt{3})^3$. 5

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3. (a) Find the absolute maximum and minimum of the following function : *5*

$$f(x) = \frac{x^3}{x+2} \quad \text{on } [-1, 1]$$

(b) Reduce the matrix $A = \begin{bmatrix} 5 & 3 & 8 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$ to normal form and hence find its rank. *5*

(c) If $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$; $\vec{b} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j} - \hat{k}$; verify that

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}.$$ *5*

(d) Find the length of function $y = 3 - 2x$ from $(0, 3)$ to $(2, -1)$ using integration. *5*

4. (a) Find the quadratic equation with real coefficients and with the following pair of roots : *5*

$$\left(\frac{m-n}{m+n}\right); \left(\frac{m+n}{m-n}\right)$$

(b) If $x = a + b$, $y = a\omega + b\omega^2$, $z = a\omega^2 + b\omega$ (where ω is a cube root of unity and $\omega \neq 1$), show that $xyz = a^3 + b^3$. *5*

(c) Solve the following system of linear equations using Cramer's rule : *5*

$$x + y = 0; \; y + z = 1; \; z + x = 3$$

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(d) If $y = ln\left[e^{x}\left(\frac{x-2}{x+2}\right)^{3/4}\right]$, find $\frac{dy}{dx}$. *5*

5. (a) A software development company took the designing and development job of a website. The designing job fetches the company ₹ 2,000 per hour and development job fetches them ₹ 1,500 per hour. The company can devote at most 20 hours per day for designing and atmost 15 hours for development of website. If total hours available for a day is at most 30, find the maximum revenue the software company can get per day. *10*

(b) Evaluate $\int x\sqrt{3-2x}\,dx$. *5*

(c) Find the vector and Cartesian equations of the line passing through the points $(-2, 0, 3)$ and $(3, 5, -2)$. *5*

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No. of Printed Pages : 6

BCS-012

BACHELOR OF COMPUTER APPLICATIONS (BCA) (REVISED)

Term-End Examination

December, 2021

BCS-012 : BASIC MATHEMATICS

Time : 3 Hours *Maximum Marks : 100*

Note : *Question number **1** is compulsory. Attempt any **three** questions from the remaining questions.*

1. (a) Find the inverse of matrix : 5

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

(b) If 7 times the 7th term of an A.P. is equal to 11 times the 11th term of the A.P., find its 18th term. 5

(c) If z is a complex number such that $|z - 2i| = |z + 2i|$, show that $\text{Im}(z) = 0$. 5

(d) Show that $|\vec{a}|\vec{b} + |\vec{b}|\vec{a}$ is perpendicular to $|\vec{a}|\vec{b} - |\vec{b}|\vec{a}$, for any two non-zero vectors $\vec{a}$ and $\vec{b}$. 5

(e) Use the principle of mathematical induction to show that : 5

$$1 + 4 + 7 + \ldots\ldots + (3k - 2) = \frac{1}{2}k(3k - 1)$$

(f) Evaluate $\int \frac{dx}{e^x + 1}$. 5

(g) Find the quadratic equation whose roots are $(2 - \sqrt{3})$ and $(2 + \sqrt{3})$. 5

(h) Find the length of the curve :

$$y = 3 + \frac{1}{2}(x)$$

from (0, 3) to (2, 4). 5

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2. (a) Find the shortest distance between : 5

$$\vec{r_1} = (1 + \lambda)\,\hat{i} + (2 - \lambda)\hat{j} + (1 + \lambda)\,\hat{k}$$

and $\vec{r_2} = 2\,(1 + \mu)\,\hat{i} + (1 - \mu)\hat{j} + (-1 + 2\mu)\,\hat{k}$

(b) Find the points of local minima and local maxima, for function : 5

$$f(x) = \frac{3}{4}x^4 - 8x^3 + \frac{45}{2}x^2 + 2015$$

(c) Find the sum of all integers between 100 and 1000 which are divisible by 7. 5

(d) If $A = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 5 & 2 \\ 2 & -1 & 7 \end{bmatrix}$, show that A is row equivalent to I_3. 5

3. (a) A stone is thrown into a lake, producing circular ripple. The radius of the ripple is increasing at the rate of 5 m/s. How fast is

the area inside the ripple increasing when the radius is 10 m ? 5

(b) If $(x + iy)^{1/3} = a + ib$, prove that : 5

$$\frac{x}{a} + \frac{y}{b} = 4\,(a^2 - b^2)$$

(c) Find the 10th term of the harmonic progression : 5

$$\frac{1}{7}, \frac{1}{15}, \frac{1}{23}, \frac{1}{31}, \ldots\ldots\ldots$$

(d) For any two vectors $\vec{a}$ and $\vec{b}$, show that : 5

$$\left|\vec{a} + \vec{b}\right| \le \left|\vec{a}\right| + \left|\vec{b}\right|$$

4. (a) Determine the values of x for which :

$$f(x) = 5x^{3/2} - 3x^{5/2},\ x > 0$$

is increasing and decreasing. 5

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(b) Solve the following system of liner equations by using matrix inverse : 10

$$3x + 4y + 7z = -2$$

$$2x - y + 3z = 6$$

$$2x + 2y - 3z = 0$$

and hance, obtain the value of $3x - 2y + z$.

(c) Find the area bounded by the curves $y = x^2$ and $y^2 = x$. 5

5. (a) If $y = \left(x + \sqrt{x^2 + 1}\right)^3$, find $\frac{dy}{dx}$. 5

(b) A company wishes to invest at most \$ 12,000 in project A and project B. Company must invest at least \$ 2,000 in project A and at least \$ 4,000 in project B.

If project A gives return of 8% and project B gives return of 10%, find how much money is to be invested in the two projects to maximize the return. 10

(c) Solve the equation :

$$2x^3 - 15x^2 + 37x - 30 = 0$$

if roots of the equation are in A. P. 5

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No. of Printed Pages : 4

BCS-012

BACHELOR OF COMPUTER APPLICATIONS (BCA) (Revised)

Term-End Examination

June, 2022

BCS-012 : BASIC MATHEMATICS

Time : 3 hours *Maximum Marks : 100*

Note : *Question number **1** is **compulsory**. Attempt any **three** questions from the remaining questions.*

1. (a) Solve the following system of linear equations using Cramer's rule : *5*

$$x + y = 0;\ \ y + z = 1;\ \ z + x = 3$$

(b) If 1, ω and ω^2 are cube roots of unity, show that

$$(2 - \omega)(2 - \omega^2)(2 - \omega^{10})(2 - \omega^{11}) = 49.$$ *5*

(c) Evaluate the integral $I = \int \frac{x^2}{(x+1)^3}\,dx$. *5*

(d) Solve the inequality $\frac{5}{|x-3|} < 7$. *5*

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(e) Show that $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (b-a)(c-a)(c-b)$. *5*

(f) Find the quadratic equation whose roots are $(2-\sqrt{3})$ and $(2+\sqrt{3})$. *5*

(g) Find the sum of an Infinite G.P., whose first term is 28 and fourth term is $\frac{4}{49}$. *5*

(h) If z is a complex number such that $|z-2i| = |z+2i|$, show that $\text{Im}(z) = 0$. *5*

2. (a) Evaluate $\lim_{x \to 0} \frac{\sqrt{1+2x} - \sqrt{1-2x}}{x}$. *5*

(b) Prove that the three medians of a triangle meet at a point called centroid of the triangle which divides each of the medians in the ratio 2 : 1. *7*

(c) A young child is flying a kite which is at a height of 50 m. The wind is carrying the kite horizontally away from the child at a speed of 6·5 m/s. How fast must the kite string be let out when the string is 130 m ? *8*

3. (a) Using Principle of Mathematical Induction, show that $n(n+1)(2n+1)$ is a multiple of 6 for every natural number n. *5*

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(b) Find the points of local minima and local maxima for

$$f(x) = \frac{3}{4}x^4 - 8x^3 + \frac{45}{2}x^2 + 2015.$$ 5

(c) Determine the 100th term of the Harmonic Progression $\frac{1}{7}, \frac{1}{15}, \frac{1}{23}, \frac{1}{31}, \ldots$ 5

(d) Find the length of the curve $y = 2x^{3/2}$ from the point (1, 2) to (4, 16). 5

4. (a) Determine the shortest distance between

$$\vec{r}_1 = (1+\lambda)\hat{i} + (2-\lambda)\hat{j} + (1+\lambda)\hat{k} \text{ and}$$

$$\vec{r}_2 = 2(1+\mu)\hat{i} + (1-\mu)\hat{j} + (-1+2\mu)\hat{k}.$$ 5

(b) Find the area lying between two curves

$$y = 3 + 2x, \quad y = 3 - x, \quad 0 \le x \le 3,$$

using integration. 5

(c) If $y = 1 + ln\,(x + \sqrt{x^2+1})$, prove that

$$(x^2+1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 0.$$ 5

(d) Find the angle between the lines

$$\vec{r}_1 = 2\hat{i} + 3\hat{j} - 4\hat{k} + t(\hat{i} - 2\hat{j} + 2\hat{k}) \text{ and}$$

$$\vec{r}_2 = 3\hat{i} - 5\hat{k} + s(3\hat{i} - 2\hat{j} + 6\hat{k}).$$ 5

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5. (a) If $A = \begin{bmatrix} -1 & 2 & 3 \\ 4 & 5 & 7 \\ 5 & 3 & 4 \end{bmatrix}$, show that A(adj A) = 0. *5*

(b) Use De-Moivre's theorem to find $(\sqrt{3} + i)^3$. *5*

(c) Show that $|\vec{a}|\,\vec{b} + |\vec{b}|\,\vec{a}$ is perpendicular to $|\vec{a}|\,\vec{b} - |\vec{b}|\,\vec{a}$, for any two non-zero vectors $\vec{a}$ and $\vec{b}$. *5*

(d) If $y = ln\left[e^x\left(\frac{x-2}{x+2}\right)^{3/4}\right]$, find $\frac{dy}{dx}$. *5*

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BCS-012

BACHELOR OF COMPUTER APPLICATIONS (BCA) (REVISED)

Term-End Examination

December, 2022

BCS-012 : BASIC MATHEMATICS

Time : 3 Hours *Maximum Marks : 100*

Note : *Question number **1** is compulsory. Attempt any **three** questions from the remaining questions.*

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1. (a) If $A = \begin{bmatrix} 3 & 4 & -5 \\ 1 & 1 & 0 \\ 1 & 1 & 5 \end{bmatrix}$, show that A is row equivalent to I_3. 5

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(b) Find the sum of an infinite G. P., whose first term is 28 and fourth term is $\frac{4}{49}$. 5

(c) Solve the inequality $\frac{5}{|x-3|} < 7$. 5

(d) Evaluate $\int \frac{x^2}{(x+2)^3}\,dx$. 5

(e) For any vectors $\vec{a}$ and $\vec{b}$, show that $\left|\vec{a}+\vec{b}\right| \le \left|\vec{a}\right| + \left|\vec{b}\right|$. 5

(f) Find the area bounded by the curves $y = x^2$ and $y^2 = x$. Also draw graph for the same. 5

(g) If z is a complex number such that $|z - 2i| = |z + 2i|$, show that Im $(z) = 0$. 5

(h) Find the quadratic equation whose roots are $(2-\sqrt{3})$ and $(2+\sqrt{3})$. 5

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2. (a) Show that : 5

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (y - x)(z - x)(z - y)$$

(b) Find $(\sqrt{3} + i)^3$ by using De Moivre's theorem. 5

(c) If $y = ax + \frac{b}{x}$, show that : 5

$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y = 0$$

(d) Find the points of discontinuity of the following function : 5

$$f(x) = \begin{cases} x^2 , & \text{if } x > 0 \\ x + 3, & \text{if } x \leq 0 \end{cases}$$

3. (a) Solve the following system of linear equations using Cramer's rule : 5

$$x + y = 0$$

$$y + z = 1$$

$$z + x = 3$$

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(b) If the first term of an A. P. is 22, the common difference is – 4, and the sum of n terms is 64, then find n. 5

(c) Find the length of the curve $y = 3 + \frac{x}{2}$ from (0, 3) to (2, 4). 5

(d) If $\vec{a}, \vec{b}, \vec{c}$ are coplanar vectors, then prove that $\vec{a} + \vec{b}, \vec{b} + \vec{c}$ and $\vec{c} + \vec{a}$ are also coplanar vectors. 5

4. (a) A child is holding string a flying kite, which is at the height of 50 m, from the ground. The wind carries away the kite horizontally, from the child, at the rate of 6.5 m/s. Determine the rate at which the kite string must be let out when the string is 130 m. 5

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(b) Using determinants, find the area of triangle whose vertices are (1, 2), (–2, 3) and (–3, –4). 5

(c) Using the principle of mathematical induction, prove that :

$$\frac{1}{(1)(2)} + \frac{1}{(2)(3)} + \ldots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for every natural number n. 5

(d) Reduce the matrix $A = \begin{bmatrix} 5 & 3 & 8 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$ to normal form and hence find its rank. 5

5. (a) Find the vector and Cartesian equations of the line passing through the points (–2, 0, 3) and (3, 5, –2). 5

(b) If $y = \ln\left[e^x \left(\frac{x-2}{x+2}\right)^{3/4}\right]$, find $\frac{dy}{dx}$. 5

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(c) A person wishes to invest at most ` 12,000 in ‘ option A’ and ‘option B’. He must invest at least ` 2,000 in ‘option A’ and at least ` 14,000 in ‘option B’. If ‘option A’ gives return of 8% and ‘option B’ gives return of 10%, determine how much investment should be done in respective options to maximize the returns. 10

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BCS-012

BACHELOR OF COMPUTER APPLICATIONS (BCA) (REVISED)

Term-End Examination

June, 2023

BCS-012 : BASIC MATHEMATICS

Time : 3 Hours *Maximum Marks : 100*

Note : *Question Number **1** is compulsory. Attempt any **three** questions from the remaining questions.*

1. (a) If $A = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$ and $(A + B)^2 = A^2 + B^2$, find a and b. 5

(b) Show that $n(n+1)(2n+1)$ is a multiple of 6 for every natural number n. 5

(c) If 1, ω and ω^2 are cube roots of unity, show that : 5

$$(2-\omega)(2-\omega^2)(2-\omega^{10})(2-\omega^{11}) = 49.$$

(d) Show that $|\vec{a}|\vec{b} + |\vec{b}|\vec{a}$ is perpendicular to $|\vec{a}|\vec{b} - |\vec{b}|\vec{a}$, for any two non-zero vectors $\vec{a}$ and $\vec{b}$. 5

(e) Solve the equation $2x^3 - 15x^2 + 37x - 30 = 0$, given that the roots of the equation are in A.P. 5

(f) Evaluate the integral : 5

$$I = \int \frac{x^2}{(x+1)^3}\, dx.$$

(g) Use first derivative test to find the local maxima and local minima if the function $f(x) = x^3 - 12x$. 5

(h) Prove that the three medians of a triangle meet at a point called centroid of the triangle which divides each of the medians in the ratio 2 : 1. 5

2. (a) Verify that $2 + 2^2 + ... + 2^n = 2^{n+1} - 2$, using the principle of mathematical induction. (Here, n represents natural numbers). 5

(b) Determine the 10th term of the Harmonic Progression $\frac{1}{7}, \frac{1}{15}, \frac{1}{23}, ...$. 5

(c) Evaluate : 5

$$\int \frac{dx}{\sqrt{x} + x}.$$

(d) Solve the following system of equations, by using Cramer's rule : 5

$$x + 2y + 2z = 3$$
$$3x - 2y + z = 4$$
$$x + y + z = 2.$$

3. (a) Given $x = a + b$, $y = a\omega + b\omega^2$, $z = a\omega^2 + b\omega$. Verify that $xyz = a^3 + b^3$. 5

(where ω is cube root of unity and $\omega \neq 1$).

(b) Given $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, perform the following : 5+5

(i) Determine A^{-1} and A^3.

(ii) Verify that $A^2 - 4A - 5I_3 = O$.

(c) If the roots of $ax^3 + bx^2 + cx + d = 0$ are in A.P., show that $2b^3 - 9abc + 27a^2d = 0$. 5

4. (a) Determine the points of local extrema of the function : 5

$$f(x) = \frac{3}{4}x^4 - 8x^3 + \frac{45}{2}x^2 + 2015.$$

(b) Calculate the shortest distance between vectors $\vec{r}_1$ and $\vec{r}_2$ given below : 5

$$\vec{r}_1 = (1 + \lambda)\hat{i} + (2 - \lambda)\hat{j} + (1 + \lambda)\hat{k}$$

and

$$\vec{r}_2 = 2(1 + \mu)\hat{i} + (+2 - \mu)\hat{j} + (-1 + 2\mu)\hat{k}$$

(c) Determine the values of x for which the function $f(x) = 5x^{3/2} - 3x^{5/2}$, $(x > 0)$ is : 5

(i) increasing

(ii) decreasing

(d) If $|z-2i| = |z+2i|$, verify that $\text{Im}(z) = 0$ (where z is a complex number). 5

5. (a) Find the direction cosines of the line passing through the two points (1, 2, 3) and (–1, 1, 0). 5

(b) Find the area bounded by the curves $y = x^2$ and $y = x$. 5

(c) Two tailors A and B earn ₹ 150 and ₹ 200 per day respectively. Tailor A can stitch 6 shirts and 4 pants while Tailor B can stitch 10 shirts and 4 pansts per day. How many days shall each work if it is desired to produce (at least) 60 shirts and 32 pants at a minimum labour cost ? Also calculate the least cost. 10

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BCS–012

BACHELOR OF COMPUTER APPLICATIONS (BCA) (REVISED)

Term-End Examination

December, 2023

BCS-012 : BASIC MATHEMATICS

Time : 3 Hours *Maximum Marks : 100*

Note : *Question Number **1** is compulsory. Attempt any **three** questions from the remaining questions.*

1. (a) Show that : 5

$$\begin{vmatrix} b-c & c-a & a-b \\ c-a & a-b & b-c \\ a-b & b-c & c-a \end{vmatrix} = 0 .$$

(b) If $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$ and $f(x) = x^2 - 4x + 7$, show that $f(A) = O_{2\times 2}$. Use this result to find A^5. 5

(c) Show that 7 divides $2^{3n} - 1 \ \forall n \in N$. 5

(d) If $1, \omega, \omega^2$ are cube roots of unity, show that : 5

$$(1 + \omega)(1 + \omega^2)(1 + \omega^3)(1 + \omega^4)(1 + \omega^6)(1 + \omega^8) = 4$$

(e) If $y = ae^{mx} + be^{-mx} + 4$, show that : 5

$$\frac{d^2y}{dx^2} = m^2(y - 4).$$

(f) If α, β are roots of $x^2 - 2kx + k^2 - 1 = 0$ and $\alpha^2 + \beta^2 = 10$, find k. 5

(g) Find the value of λ for which the vectors : 5

$$\vec{a} = 2\hat{i} - 4\hat{j} + 3\hat{k},$$

$$\vec{b} = \lambda\hat{i} - 2\hat{j} + \hat{k}$$

and $$\vec{c} = 2\hat{i} + 3\hat{j} + 3\hat{k}$$

are coplanar.

(h) Find the angle between the pair of lines : 5

$$\frac{x-5}{2} = \frac{y-3}{3} = \frac{z-1}{-3}$$

and $$\frac{x}{3} = \frac{y-1}{2} = \frac{z+5}{-3}.$$

2. (a) Solve the following set of linear equations by using matrix inverse : 5

$$3x + 4y + 7z = -2$$

$$2x - y + 3z = 6$$

$$2x + 2y - 3z = 0.$$

(b) Use the principle of mathematical induction to prove that : 5

$$1^3 + 2^3 + ... + n^3 = \frac{1}{4} n^2 (n+1)^2,$$

for every natural number n.

(c) Find how many terms of the GP $\sqrt{3}, 3, 3\sqrt{3}$, ... add up to $120 + 40\sqrt{3}$. 5

(d) Write De Moivre's theorem and use it to find $(i + \sqrt{3})^3$. 5

3. (a) If $A = \begin{bmatrix} -1 & 2 & 3 \\ 4 & 5 & 7 \\ 5 & 3 & 4 \end{bmatrix}$, show that A(adj A) = 0. 5

(b) Solve the inequality $\left|\frac{x-4}{2}\right| \leq \frac{5}{12}$ and graph the solution set. 5

(c) Solve the equation $8x^3 - 14x^2 + 7x - 1 = 0$, given that roots are in GP. 5

(d) Verify that $f(x) = 1 + x^2 \ln\left(\frac{1}{x}\right)$ has a local maxima at $x = \frac{1}{\sqrt{e}}, (x > 0)$. 5

4. (a) Evaluate : 5

$$\lim_{x \to 0} \frac{\sqrt{1+2x} - \sqrt{1-2x}}{x}.$$

(b) Find the shortest distance between the lines : 5

$$\vec{r_1} = (3\hat{i} + 4\hat{j} - 2\hat{k}) + t(-\hat{i} + 2\hat{j} + \hat{k})$$

and $\vec{r_2} = (\hat{i} - 7\hat{j} - 2\hat{k}) + t(\hat{i} + 3\hat{j} + 2\hat{k})$.

(c) Determine the length of curve $y = \frac{2}{3}x^{\frac{3}{2}}$ from (0, 0) to $\left(1, \frac{2}{3}\right)$. 5

(d) Find the sum of all the integers between 100 and 1000 that are divisible by 7. 5

5. (a) Determine the area between the two curves $y = 3 + 2x, y = 3 - x, 0 \leq x \leq 3$ using integration. 5

(b) Find the direction cosines of the lines passing through the two points (1, 2, 3) and (–1, 1, 0). 5

(c) Find the maximum value of $2a + 5b$ subject to the following constraints : 5

$$-3a - 2b \leq -6$$

$$-2a + b \leq 2$$

$$4a + 6b \leq 24$$

$$2a - 3b \leq 3$$

$$a \geq 0 \text{ and } b \geq 0.$$

(d) Reduce the matrix $A = \begin{bmatrix} 5 & 3 & 8 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$ to normal form and hence find its rank. 5

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BCS–012

BACHELOR OF COMPUTER APPLICATIONS (BCA) (REVISED)

Term-End Examination

June, 2024

BCS-012 : BASIC MATHEMATICS

Time : 3 Hours *Maximum Marks : 100*

Note : *Question No.* ***1*** *is compulsory. Attempt any* ***three*** *questions from the remaining questions.*

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1. (a) Show that : 5

$$\begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} = 2\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

(b) If : 5

$$X = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \text{ and } I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

find $(X - I_2)^2$.

(c) Find the sum upto n terms of the series : 5

$$0.3 + 0.33 + 0.333 + ...$$

(d) If 7 times the 7th term of an A. P. is equal to 11 times the 11th term of the A. P., find the 18th term of A.P. 5

(e) If $1, \omega, \omega^2$ are the cube roots of unity, find : 5

$$(2 + \omega + \omega^2)^6 + (3 + \omega + \omega^2)^6$$

(f) Find the quadratic equations with real coefficients and with the following pair of roots : 5

$$\left(\frac{m-n}{m+n}\right), -\left(\frac{m+n}{m-n}\right)$$

(g) Evaluate : 5

$$\int x\sqrt{3-2x}\, dx$$

(h) A spherical balloon is being inflated at the rate of 900 cubic centimetres per second. How fast is the radius of the balloon increasing when the initial radius is 25 cm ? 5

2. (a) Solve the following set of equations by using the matrix method : 5

$$2x - y + 3z = 5$$

$$3x + 2y - z = 7$$

$$4x + 5y - 5z = 9$$

(b) Use the principle of mathematical induction to show that : 5

$$1 + 4 + 7 + ... + (3n - 2) = \frac{1}{2} n(3n - 1)$$

(c) Find the quadratic equation with real coefficients and with the pair of roots : 5

$$\frac{1}{5 - \sqrt{72}}, \frac{1}{5 + 6\sqrt{2}}$$

(d) If :

$$\left(\frac{1 - i}{1 + i}\right)^{10} = a + ib,$$

then show that $a = 1$ and $b = 0$. 5

3. (a) If :

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 5 & 2 \\ 2 & -1 & 7 \end{bmatrix},$$

show that A is row equivalent to I_3. 5

(b) Solve the inequality :

$$\frac{2x-5}{x+2} < 5, \quad x \in R$$

Also graph the solution set. 5

(c) Solve the equation :

$$32x^3 - 48x^2 + 22x - 3 = 0$$

given the roots are in A.P. 5

(d) Determine the point of local maxima and local minima of : 5

$$f(x) = x^3 - 6x^2 + 9x + 100$$

4. (a) Check the continuity of the function $f(x)$ given below at $x = 0$: 5

$$f(x) = \begin{cases} \dfrac{2|x|}{x}; & x \neq 0 \\ 0 \quad ; & x = 0 \end{cases}$$

(b) Determine the vector and the cartesian equations of the line passing through the point (1, –1, –2) and parallel to the vector : 5

$$3\hat{i} - 2\hat{j} + 5\hat{k}$$

(c) Determine the length of the curve $y = 2x^{3/2}$ from point (1, 2) to (4, 16). 5

(d) Find the sum of all integers between 100 and 1000 that are divisible by 9. 5

5. (a) Find the maximum value of $5x + 2y$ subject to the following constraints : 5

$$-2x - 3y \leq -6$$

$$x - 2y \leq 2$$

$$6x + 4y \leq 24$$

$$-3x + 2y \leq 3$$

$$x \geq 0,\ y \geq 0.$$

(b) Find the area bounded by the curves $y = x^2$ and $y^2 = x$. 5

(c) Reduce the matrix :

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

to its normal form and hence determine its rank. 5

(d) If :

$$\vec{a} + \vec{b} + \vec{c} = 0$$

show that :

$$\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$$

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BCS–012

BACHELOR OF COMPUTER APPLICATIONS (BCA) (REVISED)

Term-End Examination

December, 2024

BCS-012 : BASIC MATHEMATICS

Time : 3 Hours *Maximum Marks : 100*

Note : *(i) Question No.* ***1*** *is compulsory.*

(ii) Attempt any ***three*** *questions from the remaining questions.*

1. (a) Show that : 5

$$\begin{vmatrix} 1 & a & a^2 \\ a^2 & 1 & a \\ a & a^2 & 1 \end{vmatrix} = (a^3 - 1)^2$$

(b) If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, show that : 5

$$A^2 - 4A - 5I_3 = O_3$$

(c) Use the principle of mathematical induction to show that $3^{2n} - 1$ is divisible by 8 for each natural number n. 5

(d) If in an A. P. $a = 2$ and sum of the first five terms is one fourth of the sum of next five terms, show that $a_{20} = -112$. 5

(e) If $z \in \mathbf{C}$, where $\mathbf{C}$ is a set of complex numbers and $|z| = 1, z \neq -1$, show that $\frac{z-1}{z+1}$ is purely imaginary. 5

(f) If two roots r_1 and r_2 of the quadratic equation $x^2 + kx + 12 = 0$ are such that $(r_1 - r_2) = 1$, find k. 5

(g) Two tailors A and B, earn ₹ 600 per day and ₹ 800 per day respectively. Tailor A can stitch 6 shirts and 4 pants per day,

while Tailor B can stitch 10 shirts and 4 pants per day. How many days shall each work if it is desired to produce (at least) 60 shirts and 32 pants at a minimum labour cost ? Also calculate the least cost. 10

2. (a) Using elementary row operations, find inverse of the matrix : 5

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

(b) How many terms of the G. P. :

$$\sqrt{3}, 3, 3\sqrt{3}, ...$$

add upto $39 + 13\sqrt{3}$? 5

(c) If $1, \omega, \omega^2$ are cube roots of unity, show that : 5

$$(2 + 3\omega + 2\omega^2)^9 \times (4 + 3\omega + 3\omega^2)^9 = 1$$

(d) If α and β are roots of the quadratic equation $3x^2 - 4x + 1 = 0$, form an equation whose roots are $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$. 5

3. (a) Find : 5

$$\lim_{x \to 0} \frac{\sqrt{x+3} - \sqrt{3}}{x}$$

(b) If $y = ae^{mx} + be^{-mx}$, show that : 5

$$\frac{d^2y}{dx^2} = m^2 y$$

(c) Determine the interval in which :

$$f(x) = -2x^3 + 24x + 7$$

is increasing. 5

(d) If a mothball evaporates at a rate proportional to its surface area $4\pi r^2$, show that its radius decreases at a constant rate. 5

4. (a) Evaluate : 5

$$\int \frac{x}{\sqrt{x+3}} dx$$

(b) Evaluate : 5

$$\int_a^b \frac{\log x}{x} dx, \quad a > b > 0$$

(c) Find the area bounded by $y = \sqrt{x}$ and $y = x$. 5

(d) If $\vec{a}$ and $\vec{b}$ are two vectors, show that : 5

$$|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$$

5. (a) Show that the lines :

$$\frac{x-5}{4} = \frac{y-7}{4} = \frac{z+3}{-5}$$

and $$\frac{x-8}{7} = \frac{y-4}{1} = \frac{z-5}{3}$$

intersect. 5

(b) Solve the inequality : 5

$$\left|\frac{x-3}{2}\right| \leq 1$$

(c) Using determinants, find the area of a triangle whose vertices are (1, 2), (–2, 3) and (–3, –4). 5

(d) If $y = \log_e \left[e^x \left(\frac{x-2}{x+2} \right)^{3/4} \right]$, find $\frac{dy}{dx}$. 5

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BCS–012

BACHELOR OF COMPUTER APPLICATIONS (BCS/BCA)

Term-End Examination

June, 2025

BCS-012 : BASIC MATHEMATICS

Time : 3 Hours *Maximum Marks : 100*

Note : *Question No.* ***1*** *is compulsory. Attempt any* ***three*** *questions from the remaining questions.*

1. (a) Show that : 5

$$\begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} = 2\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

(b) If :

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$$

find $f(A)$, where $f(x) = x^2 - 5x + 6$. 5

(c) Use the principle of mathematical induction to show that :

$$1+2+2^2+.....+2^{n-1}=2^n-1$$

for each $n \in \mathbf{N}$, where N is a set of natural numbers. 5

(d) If $|z-1|=|z-i|$, show that Re $(z)=\text{Im}(z)$. 5

(e) Find the sum to n terms of the series : 5

$$9+99+999+.....$$

(f) If α and β are roots of the equation $2x^2-8x-5=0$, find a quadratic equation whose roots are α^2 and β^2. 5

(g) Kiaan wants to buy some colour boxes and books to donate to an orphanage. He wishes to buy at least 4 books and 4 colour boxes. A colour box costs ₹ 200 whereas a book costs ₹ 400. How many colour boxes and books should he buy so that the expenditure does not exceed ₹ 4,000 and at the same time he can buy maximum number of items ? 10

2. (a) Using elementary row operations, find inverse of the matrix : 5

$$\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

(b) If $x = 3 + 2i$, find the value of $x^4 - 4x^3 + 4x^2 + 8x + 40$. 5

(c) Find the sum of an infinite G. P. whose first term is 15 and the fourth term is 3/25. 5

(d) If α and β are roots of $x^2 - 4x + 5 = 0$, find the quadratic equation whose roots are $\alpha^2 + 3$ and $\beta^2 + 3$. 5

3. (a) Find : 5

$$\lim_{x \to 0} \frac{\sqrt{x+5} - \sqrt{5}}{\sqrt{x}}$$

(b) If : 5

$$y = \left[x(x-1)(x+2) \right]^{5/7},$$

find $\dfrac{dy}{dx}$.

(c) Find the local extrema of : 5

$$f(x) = \frac{3}{4}x^4 - 8x^3 + \frac{45}{2}x^2 + 105$$

(d) If : 5

$$y = ax + \frac{b}{x},$$

show that :

$$x^2 \frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0.$$

4. (a) Evaluate : 5

$$\int x\sqrt{x+2}\,dx$$

(b) Evaluate : 5

$$\int_0^1 \frac{x}{\sqrt{1-x^2}}\,dx$$

(c) Find length of the curve $y = 4 + 3x$, from (0, 4) to (2, 10). 5

(d) Show that : 5

$$\left(\vec{a} \times \vec{b}\right).\left(\vec{c} \times \vec{d}\right) =$$

$$\left(\vec{a}.\vec{c}\right)\left(\vec{b}.\vec{d}\right) - \left(\vec{a}.\vec{d}\right)\left(\vec{b}.\vec{c}\right)$$

5. (a) Find the shortest distance between the lines : 5

$$\frac{x-1}{2}=\frac{y+1}{3}=z$$

and $$\frac{x+1}{5}=\frac{y-2}{1}=\frac{z-2}{0}.$$

(b) Solve the inequality $\left|\frac{2x-1}{3}\right| \leq 2$. 5

(c) Using determinants, show that the points A $(a, b+c)$, B$(b, c+a)$ and C $(c, a+b)$ are collinear. 5

(d) If : 5

$$y = 2e^{x} + e^{-x}$$

and $$\frac{d^2y}{dx^2} = ky,$$

find k.

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BCS–012

BACHELOR OF COMPUTER APPLICATIONS

(BCA) (REVISED)

Term-End Examination

December, 2025

BCS-012 : BASIC MATHEMATICS

Time : 3 Hours *Maximum Marks : 100*

Note : *Question No. 1 is compulsory. Attempt any* ***three*** *questions from the remaining questions.*

1. (a) Show that : 5

$$\begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix} = 4a^2b^2c^2$$

(b) If $A = \begin{pmatrix} -1 & -2 \\ 3 & 6 \end{pmatrix}$, show that $A^2 = 5A$. 5

(c) Use principle of mathematical induction to show that : 5

$$1 + 3 + \ldots\ldots\ldots + (2n - 1) = n^2, \ \forall n \in \mathbf{N}$$

(where **N** is a set of natural numbers).

(d) If $m \neq n$ and m times the mth term of an Arithmetic Progression (A.P.) is equal to n times the nth term of the A.P., then show that $(m + n)$th term of the A.P. is zero. 5

(e) If $z \in \mathbf{C}$ and $|z - i| = |z + i|$, show that $\text{Im}(z) = 0$, where z is a complex number and **C** is a set of complex numbers. 5

(f) Find the roots of the equation :

$$x^3 - 13x^2 + 15x + 189 = 0$$

given that one root exceeds the other by 2 and the roots are integers. 5

(g) Evaluate integral (I) given below : 5

$$I = \int \frac{x^3}{(x+1)^2} dx$$

(h) Show that the diagonals of a rhombus are at right angles. 5

2. (a) Find the 10th term of the Harmonic Progression (H.P.) : 5

$$\frac{1}{5}, \frac{1}{11}, \frac{1}{17}, \frac{1}{23}, \ldots\ldots\ldots$$

(b) If $x = a + b$, $y = a\omega + b\omega^2$ and $z = a\omega^2 + b\omega$, show that $xyz = a^3 + b^3$.

[Here $\omega \neq 1$ is a cube root of unity.] 5

(c) Find the direction cosines of the line joining (0, 1, – 1) and (3, 2, 1). 5

(d) If $y = \frac{e^x + e^{-x}}{e^x - e^{-x}}$, find $\frac{dy}{dx}$. 5

3. (a) If $y = \ln \left[e^x \left(\frac{x-2)}{x+2} \right)^{3/4} \right]$, show that

$$\frac{dy}{dx} = \frac{x^2 - 1}{x^2 - 4}.$$ 5

(b) Find the intervals in which the function $f(x) = 3x^{5/2} - 5x^{3/2}, x > 0$ is (i) increasing, and (ii) decreasing. 5

(c) Evaluate : 5

$$\int \frac{3^x + 2^x}{6^x} dx .$$

(d) If $\vec{a} = 4\hat{i} + \hat{j} + 3\hat{k}$ and $\vec{b} = -2\hat{i} + j - 2\hat{k}$, find a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$. 5

4. (a) If $x \in \mathbb{R}$, solve the inequality :

$$\frac{9}{x-3} < 5$$

(where $\mathbb{R}$ is a set of real numbers.) 5

(b) If $a, b, x, y \in \mathbb{R}$, $a \neq 0$ and $b \neq 0$:

$$(a - ib)(x + iy) = (a^2 + b^2)i,$$

find x and y. 5

(c) Use Cramer's rule to solve the system of equations : 5

$$2x - y + z = 4$$

$$3x - y = 5$$

$$2y - z = 1$$

(d) If : 5

$$A = \begin{pmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{pmatrix}$$

and $$B = \begin{pmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{pmatrix},$$

show that :

$$AB = BA = 6I_3.$$

Find A^{-1}.

5. (a) Use mathematical induction to show that : 5

$$1 + \frac{1}{2} + \frac{1}{2^2} + \ldots\ldots + \frac{1}{2^n} < 2 \quad \forall\, n \in \mathbf{N}.$$

(b) Find quadratic equation with real coefficients, one of its roots is $-2 + 3i$. 5

(c) If $\vec{a} = -\hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} - \hat{j} + 2\hat{k}$, find $(\vec{a} \times \vec{b}) \cdot \vec{c}$. 5

(d) Maximize : 5

$$P = 5x + 2y$$

subject to :

$$10x + 2y \geq 2100$$

$$x + \frac{1}{2}y \leq 600$$

$$y \leq 800$$

$$x \geq 0, y \geq 0.$$

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